Finite semidefinite representation of the simplest quantum correlation set

Abstract

The simplest Bell experiment allows two binary measurements on each of two separated systems. We prove that its complete quantum correlation set, including all marginal expectations and joint correlators, admits a finite semidefinite representation. Consequently, one fixed semidefinite program suffices to optimize every Bell functional in this scenario exactly. The proof reduces arbitrary-dimensional quantum realizations to real two-qubit correlations and constructs exact sums of squares through the singular limits in which measurements become compatible. The inequality coefficients and measurement parameters vary independently throughout the argument. A real-algebraic criterion converts these certificates into a finite global representation, whose matrices can be chosen with real algebraic entries.

1 Introduction

The simplest Bell experiment has two measurement choices and two outcomes at each site. Its quantum correlations depend on eight numbers: four individual outcome averages and four joint correlators. A finite semidefinite representation of this set would place every Bell inequality in the scenario on the same exact optimization domain.

Several descriptions already make this boundary accessible. The Navascués–Pironio–Acín hierarchy bounds quantum values with successively larger moment matrices [1]. The reduction to real two-qubit experiments gives a compact polynomial model [2], and the extreme correlations admit an analytical characterization [3]. A polynomial model alone leaves the semidefinite question unresolved: some compact convex semialgebraic sets have no finite semidefinite lift [4].

We prove that the complete quantum set in this experiment has a finite semidefinite representation. The key structure appears at a supporting Bell inequality. Its state matrix has three positive transverse directions, while its determinant reduces to eight angular monomials with nonnegative coefficients. At contact, those coefficients balance in the exponent plane. A few elementary identities turn that balance into exact squares.

The singular measurement limits require a further step. As measurements become compatible, both the angular terms and the couplings to the state can vanish. We retain their exact contact relations and combine complementary positive coefficients before rescaling. This produces bounded Gram matrices with positive real residue even when the Bell coefficients and the tested experiment vary independently. Scheiderer's tensor criterion then gives a finite lift [5]. The argument below develops these mechanisms; the accompanying *Complete proofs* establishes their source-dependent hypotheses and every singular boundary case.

2 The finite lift and the positivity it requires

Consider two separated quantum systems, with two binary measurements on each. Write their eight expectations as

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}), \qquad p(\alpha, \beta \mid x, y) = \frac{1 + \alpha a_x + \beta b_y + \alpha\beta E_{xy}}{4}.$$

Let $\mathcal{Q}_{222}$ be the set obtained from arbitrary finite-dimensional bipartite states and binary measurements.

Theorem 1. *There are finite integers N, p and real symmetric matrices of order N, denoted by $L_0, \dots, L_8$ and $M_1, \dots, M_p$, such that*

$$\mathcal{Q}_{222} = \left\{ q \in \mathbb{R}^8 : \ \exists y \in \mathbb{R}^p, \quad L_0 + \sum_{\nu=1}^{8} q_\nu L_\nu + \sum_{j=1}^{p} y_j M_j \succeq 0 \right\}.$$

The matrix entries can be chosen to be real algebraic numbers.

The decisive step is to preserve positivity when the Bell inequality and the experiment approach a singular contact independently. A stationary contact already determines the global angular minimum through a balanced sum of eight Laurent monomials. To obtain a finite lift, we must also express the full state quadratic form as a sum of squares with the two sets of parameters kept separate. The proof below explains this passage. Its source-specific estimates, including simultaneous vanishing coefficients and projective endpoints, are proved in the complete companion.

The two-qubit reduction gives a fixed polynomial model for the set. Two binary measurements can be dilated using one common embedding, then decomposed into common blocks of dimension at most two. In each block the observables can be written

$$A_0 = B_0 = Z, \quad A_1 = c_A Z + s_A X, \quad B_1 = c_B Z + s_B X, \qquad c_A^2 + s_A^2 = c_B^2 + s_B^2 = 1,$$

with X, Z the real Pauli matrices. Because these matrices are real, the real and imaginary parts of a complex state give a convex mixture of real states. Thus, for the polynomial expectation map Φ defined by these observables,

$$\mathcal{Q}_{222} = \operatorname{conv} \Phi(S^1 \times S^1 \times S^3) = \left\{ \sum_{i=1}^{9} w_i \Phi(z_i) : w_i \geq 0, \ \sum_i w_i = 1, \ z_i \in S^1 \times S^1 \times S^3 \right\}. \quad (1)$$

The first reduction is the usual real-qubit description of this scenario [2]; Chapter 2 proves the dilation, block decomposition and both inclusions, including commuting blocks. The nine-atom bound is Carathéodory's theorem in eight dimensions. Equation (1) makes the set compact and semialgebraic and gives a finite formula over $\mathbb{Q}$ for its extension to any real closed field.

Let P be the cone of affine functions nonnegative on $\mathcal{Q}_{222}$. Such a function is the slack of a Bell bound. For separately labelled real closed extensions K_c, K_t of $\mathbb{R}$, put $T = K_c \otimes_{\mathbb{R}} K_t$ and define

$$[f, q] = f_0 \otimes 1 + \sum_{\nu=1}^{8} f_\nu \otimes q_\nu, \qquad f \in P_{K_c}, \quad q \in (\mathcal{Q}_{222})_{K_t}. \quad (2)$$

The subscripts denote transfer of the fixed finite formulas. Scheiderer's Corollary 3.19 [5] identifies finite semidefinite representability with the following condition: for every real closed extension $K/\mathbb{R}$, every $f \in P_K$ and every $q \in (\mathcal{Q}_{222})_K$, their evaluation in two labelled copies of K is a finite sum of squares in $K \otimes_{\mathbb{R}} K$. We prove the stronger statement

$$[f, q] \in \Sigma T^2 \qquad \text{for every } K_c, K_t, \ f \in P_{K_c}, \ q \in (\mathcal{Q}_{222})_{K_t}. \quad (3)$$

No algebraic-disjointness assumption is imposed. The number of squares may depend on these choices; the criterion supplies one finite lift from the universal assertion.

3 A stationary contact balances the angular monomials

Choose $S = E_{00} - E_{01} + E_{10} + E_{11}$ and write $H = 1 - S/2$. Order its eight losing events as

weight	λ_{ev}	n	m	C	A	D	α	β
event	$(++\|01)$	$(+-\|00)$	$(+-\|11)$	$(-+\|10)$	$(-+\|00)$	$(-+\|11)$	$(--\|01)$	$(+-\|10)$.

Every valid slack which is negative at the corresponding PR vertex has the unique form

$$f = dH + 4\sum_i c_i p_i, \qquad d > 0, \quad c_i \geq 0. \tag{4}$$

Here p_i are its eight losing-event probabilities; evaluating at the PR vertex and its eight deterministic neighbours determines the coefficients. A slack nonnegative at every PR vertex is a nonnegative combination of event probabilities, whose separated square certificates are immediate.

For half-angles u, v and a homogeneous real state $\psi = (W, X, Y, Z)$, clear the positive measurement denominators and the state norm. With $D_u = 1 + u^2$, $D_v = 1 + v^2$, the resulting quadratic form $F_c = \psi^{\mathsf{T}} P_c(u, v)\psi$ is

$$\begin{aligned} F_c = {} & 4\lambda_{\mathrm{ev}} D_u (W + vX)^2 + 4nD_u D_v X^2 + 4m(vW - X + uvY - uZ)^2 \\ & + 4CD_v(uW - Y)^2 + 4AD_u D_v Y^2 + (4D + 2d)(uW + uvX - Y - vZ)^2 \\ & + (4\alpha + 2d)D_u(vY - Z)^2 + (4\beta + 2d)D_v(X + uZ)^2 - 2dD_u D_v Z^2. \end{aligned} \tag{5}$$

Chapter 2, Eq. (RFA10), gives its homogeneous angle form at all projective endpoints. Each event contributes a nonnegative amplitude square. The signed CHSH part has just one negative square, $-2dD_uD_vZ^2$. For $uv \neq 0$, the $Z = 0$ block of P_c contains the positive-definite form

$$2d\big[(uW + uvX - Y)^2 + (1 + u^2)v^2Y^2 + (1 + v^2)X^2\big]. \tag{6}$$

Hence P_c has three positive transverse directions and $P_c \succeq 0$ is equivalent to $\det P_c \geq 0$.

The determinant has a sparse structure:

$$\begin{aligned} \det P_c &= 16(1 + u^2)^2(1 + v^2)^2 u^2 v^2 L_c(u^2, v^2), \\ L_c(U, V) &= h_c + \sum_{e \in E} q_e U^{e_1} V^{e_2}, \end{aligned} \tag{7}$$

where $E = \{-1, 0, 1\}^2 \setminus \{0\}$ and every q_e is nonnegative. The eight factored coefficients are Eq. (SG6) in Chapter 4. Their signs are the reason a local stationarity condition determines global physicality.

Lemma 2 (Balanced angular positivity)**.** *Suppose $q_e, h_c, U_c, V_c \in K_c$, $q_e \geq 0$, $U_c, V_c > 0$, and $L_c = L_{c,U} = L_{c,V} = 0$ at (U_c, V_c). Then the separated evaluation of L_c belongs to ΣT^2 at every positive $U, V \in K_t$.*

Proof. Put $w_e = q_e U_c^{e_1} V_c^{e_2}$. The three contact equations are

$$h_c = -\sum_e w_e, \qquad \sum_e w_e e = 0.$$

Decompose this nonnegative balanced vector into minimal positive dependencies among the eight exponents. A minimal dependency is an opposite pair, or, up to symmetries of the square, one of

$$\{(1, 1), (-1, 0), (0, -1)\}, \qquad \{(1, 1), (-1, 1), (0, -1)\},$$

with integer weights $(1, 1, 1)$ and $(1, 1, 2)$, respectively. To obtain the decomposition, subtract from any such active dependency its largest nonnegative multiple. At least one active coefficient disappears, so this finishes after finitely many steps.

Set $X = U_c^{-1} \otimes U$ and $Y = V_c^{-1} \otimes V$. Each dependency now gives positive separated monomials with weighted product one. Their nonnegative differences are sums of squares by the identities

$$\begin{aligned} p + p^{-1} - 2 &= (\sqrt{p} - p^{-1/2})^2, \\ a^3 + b^3 + c^3 - 3abc &= \tfrac{1}{2}(a + b + c)\big[(a - b)^2 + (b - c)^2 + (c - a)^2\big], \\ a^4 + b^4 + c^4 + d^4 - 4abcd &= (a^2 - b^2)^2 + (c^2 - d^2)^2 + 2(ab - cd)^2. \end{aligned}$$

Use cube roots for weights $(1,1,1)$, and fourth roots with the last monomial repeated for weights $(1,1,2)$. Every root is taken separately in the source and testing fields. Expand each positive monomial prefactor before taking its square root. Multiplying by the positive source weights and summing proves the assertion. □

In one field, Lemma 2 and Eq. (6) show that a nonnegative event source is physical whenever it has a noncommuting stationary zero:

$$P_c(u_c, v_c)\psi_c = 0, \qquad \psi_c^{\mathsf{T}} P_{c,u}(u_c, v_c)\psi_c = \psi_c^{\mathsf{T}} P_{c,v}(u_c, v_c)\psi_c = 0. \tag{8}$$

The positive transverse block makes its adjugate a positive multiple of $\psi_c\psi_c^{\mathsf{T}}$, so these equations give the three scalar contact equations used in the lemma.

This also reduces the supporting inequalities. Normalize $d + \sum_i c_i = 1$. Call the angular sum coercive when zero lies in the interior of the convex hull of its active exponents. Over $\mathbb{R}$, it then diverges whenever $(\log U, \log V)$ leaves every bounded set. A coercive extreme source must attain zero; otherwise opposite small perturbations remain physical. At a fixed zero, Eq. (8) is a system of six homogeneous linear equations in the source coefficients. Extremality therefore leaves at most six positive event weights. The compatible omissions in the eight-event cycle leave at least five. Up to the CHSH symmetries, the resulting patterns are precisely

$$\begin{aligned} &\mathsf{C}_1: \quad \lambda_{\mathrm{ev}} = n = 0, \quad d, m, C, A, \alpha > 0, \quad D, \beta \geq 0, \quad D + \beta > 0, \\ &\mathsf{C}_2: \quad \lambda_{\mathrm{ev}} = \beta = 0, \quad d, m, C, A, D, \alpha > 0, \quad n \geq 0. \end{aligned} \tag{9}$$

Noncoercive sources belong to two closed families with explicit small lifts. Chapter 4 proves the extreme-source reduction, the compatibility assertion and these closed-family lifts. Chapter 5 proves that the actual contacts have the required signs and positive native gaps, including $D = 0$, $\beta = 0$ and $n = 0$. Thus no positive infinitesimal event is discarded by replacing it with zero.

4 Recovering the full state polynomial at singular contacts

The angular lemma certifies the determinant. It does not yet certify the full state polynomial in the separated ring. Ordinary state elimination can introduce an inverse depending on both the source and the test; such an inverse is not licensed in T. The local argument keeps the state amplitudes in explicit polynomial identities and controls the coefficients of their Gram matrix.

Let B_c, B_t be the bounded elements of the two fields, and put $B = B_c \otimes B_t$. The residue $\overline{a}$ of a bounded tensor is obtained by taking the real standard part in each factor.

Lemma 3 (A strict real Gram matrix)**.** *If $G \in \mathrm{Sym}_r(B)$ and $\overline{G} \succ 0$, then $z^{\mathsf{T}} G z \in \Sigma T^2$ for every $z \in T^r$. The entries of z need not be bounded or independent.*

Proof. First suppose $a \in B$ has positive real residue a_0. Write $a = a_0 + \sum_j x_j \otimes y_j$, with bounded factors and an infinitesimal factor in each summand. Distribute a_0 into positive real numbers a_j. If x_j is infinitesimal, choose a real $M_j > |y_j|$ and use

$$a_j + x_j \otimes y_j = \frac{1}{2}\left[(a_j/M_j + x_j) \otimes (M_j + y_j) + (a_j/M_j - x_j) \otimes (M_j - y_j)\right].$$

Each pure factor on the right is positive and has a square root in its own field. Interchange the factors when y_j is infinitesimal. Thus a is a tensor sum of squares.

A real congruence reduces the matrix statement to $\overline{G} = I$. Choose real $\delta > 0$ with $(r-1)\delta < 1$. Writing e_i for the coordinate vectors, the exact decomposition

$$\begin{aligned} G = &\sum_i [G_{ii} - (r-1)\delta] e_i e_i^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta + G_{ij}}{2} (e_i + e_j)(e_i + e_j)^{\mathsf{T}} \\ &+ \sum_{i<j} \frac{\delta - G_{ij}}{2} (e_i - e_j)(e_i - e_j)^{\mathsf{T}} \end{aligned}$$

has scalar coefficients of positive residue. Apply the scalar argument and pair the resulting squares with z. □

Every local certificate reduces to this lemma. If Π is a sum of positive event squares and $s = s_0 + A\xi$ is the chosen state substitution, quadratic state degree gives the exact identity

$$\begin{aligned} \mathcal{R}(\xi) := (F - \Pi)(s_0 + A\xi) &= g_0 + \sum_i g_i \xi_i + \xi^{\mathsf{T}} H \xi, \\ g_0 = [1]\mathcal{R}, \qquad g_i = [\xi_i]\mathcal{R}, \qquad H_{ij} &= \tfrac{1}{2} \partial_{\xi_i} \partial_{\xi_j} \mathcal{R}. \end{aligned} \tag{10}$$

The scalar g_0, every first state-normal coefficient g_i and the whole state Hessian H must be included. Monic polynomial division at the contact expresses their vanishing factors in the angular roots. The raw monomial support and the source bounds then show that the resulting coefficient corrections have zero residue. Contact vanishing alone does not provide either assertion. The exact restricted division theorem is in Section 6.2; the common polynomial coefficient operations are in Section B.1.1.

A useful uniform step removes a choice between a vanishing state coefficient and a vanishing angular coefficient.

Lemma 4 (Complementary source weights)**.** *Let $a, b \geq 0$ be source coefficients with $S = a + b$ a positive bounded unit. Suppose $\ell, c_i \in B$ have zero residue and*

$$\sqrt{b}\, \ell\Gamma = \sum_i c_i z_i.$$

Then $\ell\Gamma$ is a linear combination, with zero-residue bounded coefficients, of $\sqrt{a}\Gamma, z_1, \ldots, z_m$. This includes either weight equal to zero.

Proof. Use the exact identity

$$\ell\Gamma = \frac{\sqrt{a}\, \ell}{S} (\sqrt{a}\Gamma) + \frac{\sqrt{b}}{S} \sum_i c_i z_i.$$

Only the positive source unit S is inverted. Both sets of coefficients are bounded with zero residue. □

Here is a small polynomial example of the conductor step. Let $k > 0$ be a source infinitesimal, $s > 0$ a bounded source element, and $u, v > 0$ independent testing elements. Put

$$U_1 = u^2 v, \quad U_2 = kuv^2/s, \quad U_3 = k^2 s, \quad g = kuv, \qquad d_i = U_i - g.$$

Then $U_1 U_2 U_3 = g^3$. With $p_i = U_i^{1/3}$,

$$\sum_i U_i^2 - 3g^2 = \sum_i d_i^2 + g(p_1 + p_2 + p_3) \sum_{i<j} (p_i - p_j)^2 \in \Sigma T^2.$$

Thus the angular identity supplies three unit diagonal coefficients on d_1, d_2, d_3. The exact polynomial relations

$$g(u-k) = kd_1, \qquad g(v-s) = sd_2 \tag{11}$$

express contact factors in those same roots, with no division by $u-k$ or $v-s$. The first coefficient is infinitesimal. If s is a unit, the second becomes a zero-residue correction only when the complete state-normal term supplies an additional small coefficient. That coefficient bound is part of the full state-normal calculation; stationarity alone does not supply it.

The C1 application, Theorem B.23, combines polynomial conductors with complementary source weights. Its statement gives the exact source domain, and its proof verifies the complete state-normal bounds uniformly over that domain. It covers all infinitesimal independent u, w, every state ratio, and both $D = 0$ and $\beta = 0$; separate complementary results supply the unit-Alice and other projective tests.

The C2 construction uses the same operations on the full state polynomial: positive event squares remain whole, polynomial contact relations produce angular roots, and a strict real matrix controls the remaining directions. Its shared local and complementary identities are Sections C.3.1 and C.3.2; the bounded-ratio theorem is Section C.11.6. The hypotheses of these statements are proved from the original source coefficients. They are not consequences of the determinant identity alone.

Proposition 5 (Complete canonical certificate)**.** *Let K_c, K_t be any separately labelled real closed extensions of $\mathbb{R}$. For a source in either family* (9) *with an actual finite noncommuting stationary zero* (8)*, the complete homogeneous parent belongs to ΣT^2 at every independent projective testing pair and every homogeneous real or complex state.*

Proof. The full source-dependent proof is in the complete companion. After normalizing $d = 1$, the source ratio

$$\delta = \frac{d}{d + \lambda_{\mathrm{ev}} + n + m + C + A + D + \alpha + \beta}$$

is either a positive unit or a positive infinitesimal. A unit gives bounded source coefficients and is covered by Chapter A. The infinitesimal case is covered by the C1 and C2 source theorems in Chapters B and C. Those chapters prove the source partitions and the coefficient estimates needed in Eq. (10), and give all complementary testing and state charts. The noncommuting-residue theorem (Theorem 6.2) applies at arbitrary source scale. Literal commuting tests have a positive CHSH slack directly. The homogeneous parent includes both zero and infinite half-angles. The proofs retain the permitted literal $D = 0$, $\beta = 0$ and $n = 0$ faces. Applying the real result separately to the real and imaginary state parts gives the complex-state result. □

5 Assembly and the extraction of an exact pencil

The source reduction in Chapter 4 expresses any valid slack as a nonnegative combination of at most nine event, closed-family or stationary source slacks. Its fixed finite formula transfers to K_c. Equation (1) gives at most nine testing atoms in K_t. Events, the explicit closed-family lifts and Proposition 5 certify every source/test atom pair. If

$$f = \sum_j a_j f_j, \qquad q = \sum_i w_i q_i, \qquad a_j, w_i \geq 0, \quad \sum_i w_i = 1, \quad [f_j, q_i] = \sum_k h_{ijk}^2,$$

then exactly

$$[f, q] = \sum_{i,j,k} \left[(\sqrt{a_j} \otimes \sqrt{w_i}) h_{ijk} \right]^2.$$

The normalized testing weights preserve the affine constant. All positive source and testing normalizations are restored by square roots in their own fields, as in Eq. (RFA13) of Chapter 2.

This proves Eq. (3). Scheiderer's criterion now gives one finite semidefinite representation of all eight coordinates.

Finally, at any fixed N, p, equality between the proposed projection and Eq. (1) is a first-order condition over $\mathbb{Q}$ on the matrix entries: positive semidefiniteness is expressed by its principal minors. A real solution therefore has a solution in the real algebraic field. This proves the algebraic-coefficient assertion in Theorem 1. It also gives an exact extraction procedure in principle: enumerate the finite size pairs, decide each existence sentence by quantifier elimination, and return an algebraic sample for the first successful pair. The theorem ensures termination. No successful full-set pencil or useful size bound has been evaluated here; finding a small explicit representation requires further constructive work.

6 Discussion

For the two-dimensional tilted-CHSH projection, an exact matrix representation has a particularly simple form. With $t = a_0$ and $s = E_{00} + E_{01} + E_{10} - E_{11}$, that projection is

$$|t| \le 1, \qquad t^2 + s^2/4 \le 2,$$

represented by

$$\begin{pmatrix} \sqrt{2}+t & s/2 \\ s/2 & \sqrt{2}-t \end{pmatrix} \oplus (1-t) \oplus (1+t) \succeq 0.$$

The matching Bell bound and two-qubit realization establish both inclusions [7]; they are derived in the accompanying *Explicit semidefinite representations and local certificates for Q222*.

For all eight coordinates, the existence proof supplies finite matrices with real algebraic entries. Evaluating a useful such pencil requires a common finite collection of functions for the local certificates. The singular identities identify the functions that must survive this construction. Once extracted, the same feasible region will optimize every Bell functional in the scenario exactly.

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